

Particle-in-Cell Modeling of Omega Experiments on Ablation of Plasmas

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Abstract—The large electric fields due to charge separation generated by a laser pulse propagating in a plasma can accelerate the electrons to very high energies. The hydrodynamics models that are based on the local thermodynamic equilibrium assuming a Maxwellian distribution of velocities cannot describe the evolution of groups of hot electrons of different energy ranges and kinetic particle-in-cell (PIC) simulations are required. The SMILEI PIC code is used to investigate the generation of hot electrons in plasmas produced by laser pulses in the Omega extended performance (EP) experiments. In 2-D PIC simulation using SMILEI code, a Gaussian-shaped laser pulse of 100-ps duration with the intensity of 10^{15} W/cm² and a wavelength of 351 nm impinges onto a plasma with the electron density and temperature profiles. The electromagnetic field, electron, and total energies of a plasma as a function of time are evaluated. The evolution of number density, electromagnetic fields, and energy spectra of electrons is studied. The time evolution of hot electrons in a plasma is investigated. These PIC results provide insights into the contribution and importance of the kinetic effects in the expansion dynamics of ablated plasmas, thereby helping to analyze and explain the Omega experimental data and computational fluid dynamics results.

Index Terms—Dense plasma, energy spectra of hot electrons, kinetic effects, laser ablation, particle-in-cell (PIC) simulations, SMILEI PIC code.

I. INTRODUCTION

HIGH-ALTITUDE nuclear detonation can radiate 70% to 80% of its released energy as a high-intensity X-ray pulse [1]. A major effect of prompt cold X-rays (~ 1 keV) to a few microns of satellite surface materials is thermomechanical shock (TMS) induction, surface ablation, blow-off, and generation of warm dense plasma (WDP) [2], [3]. It causes damage to exposed optics, sensors, and solar cells on satellites. Solar cells are more susceptible to X-rays, since the large surface area is exposed to radiation and cannot be substantially shielded [1]. The dense surface plasmas can couple the solar cells to each other and to dielectric structures causing them to be destroyed. The physics mechanisms of formation and expansion phases of produced WDP and its thermophysical, electrical, and optical properties are basic science questions [4], [5], [6], [7].

Manuscript received 18 August 2022; revised 18 October 2022; accepted 1 November 2022. Date of publication 16 November 2022; date of current version 21 April 2023. This work was supported by the Defense Threat Reduction Agency under Grant HDTRA1-19-1-0019 and Grant HDTRA1-20-2-0001. The review of this article was arranged by Senior Editor S. J. Gitomer.

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TPS.2022.3220184>.

Digital Object Identifier 10.1109/TPS.2022.3220184

The characterization of X-ray pulse effects on the materials is complicated by the availability of very intense X-ray sources (comparable to high-intensity X-rays with a flux of $\sim 10^{28}$ – 10^{35} ph/(cm²·s) produced by a nuclear event) in the laboratory conditions and requirements of a very high level of vacuum in the chamber. The absorption of X-rays in solar array materials, formation, and spatiotemporal evolution of WDPs can be studied nowadays using a combination of high-fidelity computational models [8], [9], [10]. Modeling can be also vital to guide our fundamental understanding of the underlining physics of WDPs [11], [12], [13], [14]. In contrast, the blow-off and TMSs in the materials somewhat mimicking X-ray blow-off in nuclear environments can be produced by using the high-power laser pulses. High-power Omega extended performance (EP) laser can be used as a surrogate for ablating layers of materials and mimicking TMS and blow-off induced by X-rays [15], [16]. In contrast, the radiation hydrodynamics modeling using the Flash code [17], [18] can help to understand the role of laser pulse length on the ablation and TMS formation.

For laser intensities on the order of 10^{15} W/cm², the absorption mechanisms of laser light by the electrons are normal and anomalous skin effects and inverse bremsstrahlung process. In this nonrelativistic regime, the energy transfer to plasma ions is due to the electron–ion collisions. For long laser pulses on picosecond (ps) and nanosecond (ns) timescales, many collisions take place in a plasma that results in its very efficient collisional heating. Even for a relatively weak laser intensity (dimensionless amplitude $a_0 = 0.05$), it was pointed out that the laser pulse can generate a charge separation, induce linear wake oscillations, and lead to the ponderomotive acceleration of electrons [19]. The laser-induced plasma instabilities can also lead to the heating and accelerating of the electrons. The generation of hot (nonthermal) electrons due to the stimulated Raman scattering (SRS) and two plasmon decay (TPD) instabilities was studied using the particle-in-cell (PIC) modeling of laser–plasma interactions at a relevant laser intensity of 10^{15} W/cm² [20]. A non-Maxwellian tail of hot electrons was observed in the energy distribution. The so-called cogeneration mechanism of hot (suprathermal) electrons in an under-dense hydrocarbon plasma was recently investigated using the 2-D and 3-D PIC simulation [21]. According to this mechanism, the SRS plasma waves can generate the hot electrons which are accelerated by the TPD plasma waves. The linearly and circularly polarized lasers at normal incidence with a wavelength of 351 nm and an intensity of 1.04×10^{15} W/cm² were used. This 3-D PIC simulation

is highly relevant, since the same laser parameters are used in the present study. However, Cao et al. [21] used a constant temperature of 2 keV and they implemented a linear profile of electron density n_e within the range $0.05n_{ec} < n_e < 0.25n_{ec}$, where n_{ec} is the critical electron density defined in Section III. These plasma conditions are different from those used in the present research. It was found that the phase energy of different plasma waves can range from 32.5 keV to more than 114 keV [21]. These waves interact and accelerate the electrons. It is predicted that the hot electrons, defined in this study as those with the energy higher than 50 keV, are generated by the SRS and TPD plasma waves. Unfortunately, the energy spectra of hot electrons were not reported in this PIC study.

The goal of the present research is the PIC simulation of the experiments carried out on the Omega EP facility [15], [16] in order to provide insights into the generation of energetic (hot) electrons in under-dense and over-dense plasmas produced by laser pulses [22]. The particular interest is the time evolution of energy spectra of hot electrons. In Section II, the PIC model and SMILEI computer code are described. The PIC results are presented and discussed in Section III. Finally, the conclusion is given in Section IV.

II. PIC MODEL AND SMILEI CODE

The PIC simulation codes have become standard tools in many areas of plasma physics research [23], [24], [25], [26], [27], [28], [29]. Nowadays, the intense laser–plasma interactions can be reliably modeled using the PIC approach [30]. The results of laser experiments can be predicted and the advanced modeling of such experiments can be also accomplished. The progress and advances in the modeling approaches for performing the state-of-the-art simulations of laser pulse interactions with solids and plasmas are recently reviewed [31]. The evolution of plasma quantities and variables of interest in the PIC simulation is studied on spatial and temporal scales. A standard PIC method is appropriate for modeling collective interactions in a collisionless approximation that is suitable for weakly collisional or ideal plasmas. The close binary collisions are effectively omitted, but may be included using Monte Carlo algorithms [32]. In PIC algorithm, the mesh is introduced within the computational domain. The Debye length in a plasma should be typically resolved in order to accurately calculate the charge density. The basic PIC loop consists of the following steps [32]: 1) charges and currents created by macro-particles are interpolated and computed at the grid points; 2) Maxwell’s equations are solved on the grid to calculate the electric and magnetic fields at grid points; 3) the calculated fields are interpolated to macro-particle positions to evaluate the forces acting on macro-particles; and 4) Newton–Lorentz equations of motion are solved to calculate positions and velocities of particles at the new time step. In this way, the PIC method solves the Vlasov–Maxwell system of equations describing the evolution of a collisionless plasma. The various conditions, such as periodic, reflective, open, and thermalize, on the boundaries of computational domain can be used. Collisions, electron ionization, emission or absorption of

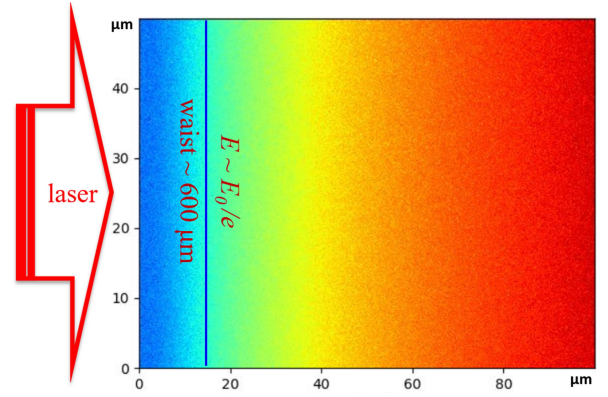


Fig. 1. Two-dimensional setup of PIC simulation where a laser interacts with a plasma having the gradients of electron density and temperature.

radiation, and other atomic processes that play an important role in laser–plasma interactions are implemented as separate models in many modern PIC codes.

The SMILEI code [33] is used to perform the PIC simulations of laser–plasma interactions. SMILEI is a well-established open-source PIC code that includes many advanced features, numerical schemes, and capabilities. The general algorithm implements the standard PIC loop and the associated numerical methods for projection of charge and current densities of macro-particles onto the grid, time and space discretization of the electromagnetic fields, solving Maxwell’s equations, and Boris macro-particle pusher. In addition, SMILEI includes many different physics modules such as the electric field and current density filters, user-defined currents (antennas), binary collisions, field, and collisional ionization. Python-based interface is used in an input file for the initialization of PIC simulations that allows to process the arithmetic operations, perform the specific physics calculations, and pass python functions (for example, electron density and temperature profiles) as arguments to SMILEI. Data collection and analysis of scalars, entire grid fields, fields at specific probes, energy spectra at screens, macro-particle trajectories, and distributions can be performed at user-defined times during the PIC simulation. A hybrid MPI-OpenMP parallelization relying on a patch-based decomposition allows for dynamic load balancing and high-performance computing on massively parallel supercomputers. The development of SMILEI was motivated by advances in laser technology. The full electromagnetic wave, temporal and spatial wave envelope, and time-space laser envelope models are available for modeling the laser light propagation in plasmas.

III. RESULTS AND DISCUSSION

The setup and results of 2D3V PIC simulation that is relevant to experiments at the Omega EP laser facility [15], [16] are presented in this section.

A. Setup of PIC Simulation

The plasma is composed of electrons and silicon (Si) ions. The setup for the 2D3V PIC simulation is shown in Fig. 1. The size of the computational domain is $100 \times 50 \mu\text{m}$. The cell size is $\Delta x = \Delta y = 21.7 \text{ nm}$ with a total number

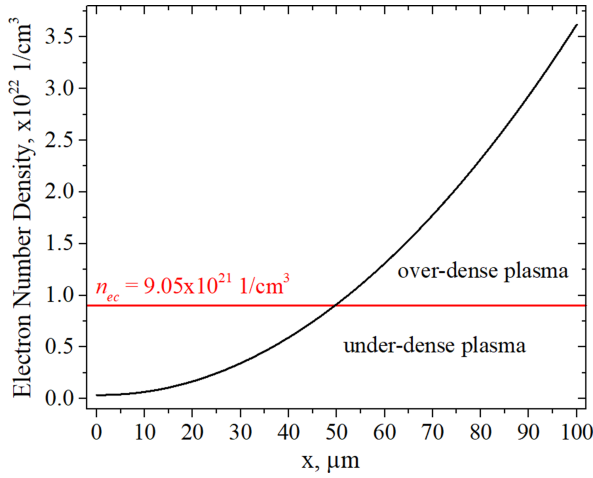


Fig. 2. Profile of electron number density along the depth of Si plasma.

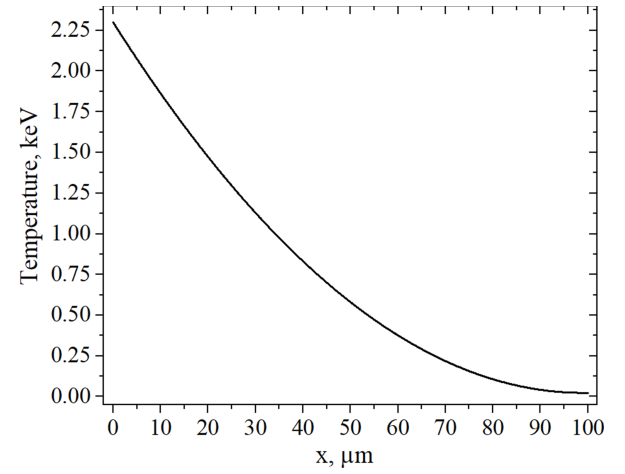


Fig. 3. Profile of electron temperature along the depth of Si plasma.

of cells 4608×2304 . The fourth-order (five points stencil) shape spline function of macro-particles was used in this PIC simulation that does not require to resolve the Debye length [32], [34]. A total number of 42 467 328 electron macro-particles and 42 467 328 Si macro-particles is placed in the domain. The simulation time is 120 ps with a timestep of 0.04863 fs (total 2467 776 timesteps). The laser pulse with linear polarization and wavelength of 351 nm (laser frequency $\omega_l = 5.37 \times 10^{15} \text{ s}^{-1}$) has a peak intensity of 10^{15} W/cm^2 (dimensionless amplitude $a_0 = 0.009435$) and propagates in the x -direction. The laser period of 1.17 fs is well resolved with 24 timesteps. The focal spot is located at $x = 0 \text{ μm}$ and $y = 25 \text{ μm}$. In the Omega EP experiments, the beam waist [y -coordinate at which the electric field E is at $1/e$ of its maximum value E_0 (shown in Fig. 1)] of laser is $\sim 600 \text{ μm}$ that is much larger than the y -dimension of the computational domain. The transverse domain size is much smaller than the lateral size of laser beam. Therefore, the laser field has nearly a constant value along the y -coordinate. The laser pulse has a Gaussian temporal profile with a 100-ps full-width-at-half-maximum and arrives at normal incidence to the plasma surface.

The initial profile of electron density along the plasma depth (x -coordinate) is shown in Fig. 2. The color map of electron density gradient in the computational domain is also illustrated in Fig. 1. There are regions of under-dense plasma localized at $x < 50 \text{ μm}$ and over-dense plasma located at $x > 50 \text{ μm}$. The red line in Fig. 2 demonstrates the value of critical electron density calculated using

$$n_{ec} = \frac{m_e \omega_l^2}{4\pi e^2}$$

where m_e is the mass of electron and e is the electronic unit charge. The density of electrons is relatively low $\sim 0.04n_{ec} \approx 3.62 \times 10^{20} \text{ cm}^{-3}$ at $x = 0 \text{ μm}$ and it increases to $\sim 4n_{ec} \approx 3.62 \times 10^{22} \text{ cm}^{-3}$ at $x = 100 \text{ μm}$. This nonlinear density profile covers both the under-dense and over-dense plasma regions that is different from a linear density profile used in [21] where only an under-dense plasma with the highest value of $n_e = 0.25n_{ec}$ was considered.

The profile of temperature along the plasma depth is shown in Fig. 3. The temperature is 2.3 keV at the front of plasma $x = 0 \text{ μm}$ and it drops along the depth of plasma reaching a room temperature at $x = 100 \text{ μm}$. To remind the reader, a constant temperature of 2 keV was used in [21].

The position, velocity, and weight functions of each electron and Si macro-particles in the computational domain are initialized based on the temperature and density profiles (see Figs. 2 and 3) which are relevant to the Omega EP laser experiments [15], [16]. The Si ions are treated as mobile. The collisions between the electrons and ions are taken into account in the PIC simulation. A collision module implemented in SMILEI is based on the Nanbu's [35] and Nanbu and Yonemura's [36] approach with the addition of relativistic particles, correction to the collision rate at low temperature, and variable Coulomb logarithm [37]. The Silver-Müller boundary condition (BC) is used for the electromagnetic fields [38] and periodic BCs are employed for macro-particles. The Silver-Müller BC is used for injecting and absorbing a laser wave from the left-hand side into the computational domain (see Fig. 1).

B. Analysis of PIC Results

The time evolution of electromagnetic, kinetic electron, and total energies in the computational domain is shown in Fig. 4. Due to the 2D3V geometry used in the PIC simulation, the energy is calculated in units MJ/cm assuming per 1-cm z -dimension. The energy of electromagnetic field sharply increases with time up to $\sim 20 \text{ ps}$ as the laser propagates into the domain and transfers energy to the electrons. After $\sim 20 \text{ ps}$, the field energy nearly saturates. This temporal profile of the integrated field energy is very similar to that reported in [21] for simulation time of 6 ps. The kinetic energy of the electrons increases by four orders of magnitude during $\sim 20 \text{ ps}$ and then becomes constant.

The mechanism of energy gain by the electrons and their acceleration is due to the ponderomotive force [39] and plasma instabilities involving nonlinear wave-electron interactions [20], [21]. The electrons are pushed away from the propagation direction of laser pulse in an under-dense plasma.

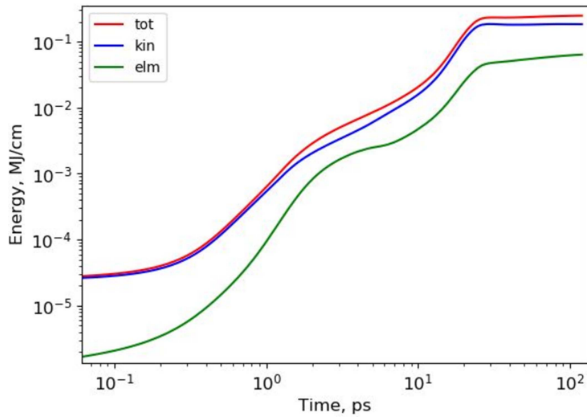


Fig. 4. Integrated electromagnetic field (elm), electron (kin), and total (tot) energies as a function of time throughout the PIC simulation.

Much heavier Si ions are also pushed in the same direction by the ponderomotive force, but they respond slower compared to electrons that leads to a charge separation. The large longitudinal electric fields are generated over a short time in very small distances (locally) due to the charge separation accelerating the electrons to high energies. Once the laser optical cycle passes through this local area, the electrons get back to their positions due to the large electrostatic attractive force created by the ions. The next laser cycles will cause the oscillation of electrons at their positions. The kinetic energy increases with time up to ~ 20 ps, as shown in Fig. 4. After that time, the saturation in laser light absorption by electrons is reached and their kinetic energy becomes constant.

The experimental studies have demonstrated that SRS and TPD instabilities induced by a 350-nm laser with intensity of 10^{15} W/cm² in an under-dense plasma can accelerate the electrons in the energy range between 35 and 350 keV [40]. The PIC simulation for a 351-nm laser with intensity of 3×10^{15} W/cm² has predicted the generation of hot electrons from plasma waves excited by SRS and accelerated up to 1 MeV [41]. The SRS and TPD instabilities occur in the plasma region at and below $0.25n_{ec} \approx 2.26 \times 10^{21}$ cm⁻³, respectively. This is a part of the plasma in Fig. 2 localized below $x \approx 23.8 \mu\text{m}$. The growth rate of SRS and TPD instabilities is governed by the gradients of electric field energy densities as demonstrated in [21] and [42]. The field energy shown in Fig. 4 increases by several orders of magnitude during ~ 20 ps, thus exciting the growth of SRS and TPD instabilities and their interaction with the electrons. These parametric SRS and TPD instabilities and their wave interactions with plasma electrons requiring a full treatment of the electromagnetic fields and the nonthermal electron energy distribution cannot be captured by the hydrodynamic approach.

The analysis of the evolution of electron number density in time within the 2-D simulation domain (see Fig. 5) revealed that the density fluctuates during ~ 120 ps around a predefined, initial density profile of electrons illustrated in Fig. 2. The electron density is $\sim 4n_{ec} \approx 3.62 \times 10^{22}$ cm⁻³ (over-dense plasma) in the region near the right edge of domain ($x > 90 \mu\text{m}$) according to a color bar in Fig. 5. However, there are very tiny local spots created by a laser

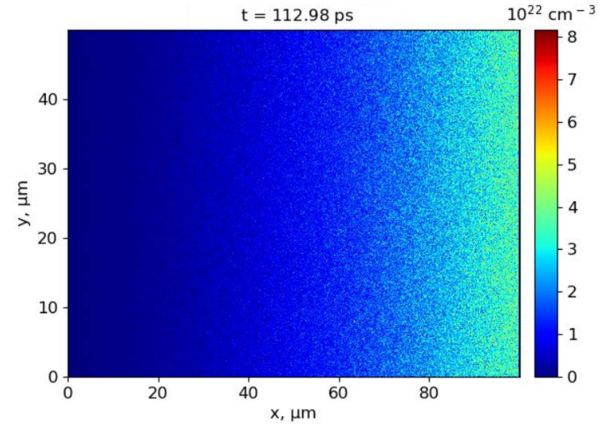


Fig. 5. Color map of number density of electrons in the computational domain at ~ 112.98 ps.

pulse where the density of electrons is two-to-three time higher ($> 8 \times 10^{22}$ cm⁻³).

The time evolution of the electric and magnetic field components within the computational domain was investigated. The electromagnetic fields evolve and increase in the strength reaching a steady state after ~ 20 ps. The color maps of longitudinal and transverse electric field components E_x and E_y as well as the z -component of magnetic field B_z are shown in Fig. 6 at 100.07 ps. It is seen in Fig. 6(a) that the positive component of E_x reaches a high value of ~ 2.5 TV/cm locally at some tiny spots due to the electron-ion charge separation. The electrons are forward accelerated by this large longitudinal electric field to high energies. There is a small negative component of E_x within the front region of a plasma. The distribution of E_x is uniform along the y -coordinate. After reaching the equilibrium state, both the positive and negative parts of transverse component of electric field E_y fluctuate around $\sim \pm 400$ GV/cm [see Fig. 6(b)]. The spatial distribution of E_y is uniform within the entire domain. Both the positive and negative parts of magnetic field component B_z increase during ~ 20 ps and then fluctuate around $\sim \pm 200$ kT [see Fig. 6(c)]. However, the spatial distribution of B_z is nonuniform within the computational domain. There are well-defined regions (domains) with negatively and positively directed components of B_z . These magnetized domains are not steady, but they change and evolve with time.

The electron energy distribution functions (EEDF) from 2-D PIC simulations are shown in Fig. 7. The spectra are illustrated at several successive times (0, 1.17, 10.05, 20.09, 23.09, and 100.07 ps). They represent all electrons in the computational domain and demonstrate a number of electrons per unit energy and unit volume as a function of the electron energy. It is seen that the plasma electrons have a broad distribution of their energies over several orders of magnitude. The electrons in this broad energy range are spread within the entire computational domain. The energy spectra of electrons shown in Fig. 7 are very similar in shapes at different times. They basically comprise three parts: the low-energy electrons, the peak with the energetic electrons, and the tail with high-energy electrons. The spectrum is shifted toward the higher energies with time as the laser light propagates into the plasma and heats the electrons (see Fig. 4). The population of hot

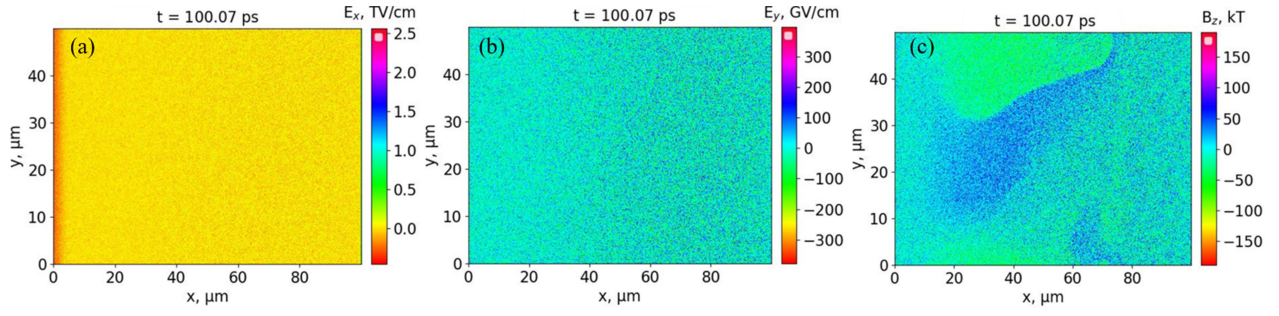


Fig. 6. Color maps of (a) x -component (longitudinal) of electric field E_x , (b) y -component (transverse) of electric field E_y , and (c) z -component of magnetic field B_z at 100.07 ps.

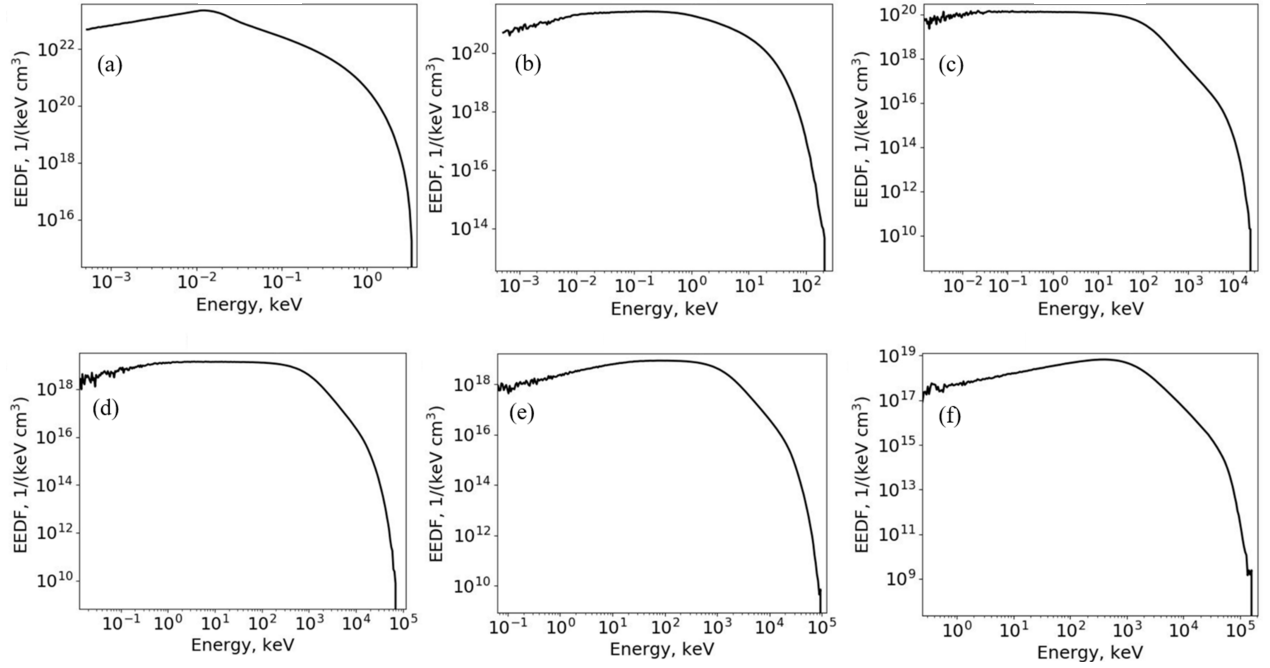


Fig. 7. Energy spectra of all electrons in the computational domain for different time moments. (a) $t = 0$ ps. (b) $t = 1.17$ ps. (c) $t = 10.05$ ps. (d) $t = 20.09$ ps. (e) $t = 23.09$ ps. (f) $t = 100.07$ ps.

(suprathermal) electrons can be defined as the number of all electrons with energies around the peak and in the tail of energy spectrum. Thus, the number of hot electrons can be characterized by the position of the peak in their energy spectrum. A relatively small portion of very energetic electrons can be found in the tail of energy distribution.

The initial spectrum of electron's kinetic energy in the computational domain [see Fig. 7(a)] is governed by the temperature (see Fig. 3) as well as number density (see Fig. 2) distributions. At $t = 0$ ps, there is only a small portion of high-energy electrons (>0.1 keV) in the tail of distribution function. The narrow energy peak near ~ 12 eV [see Fig. 7(a)] becomes much broader and shifted to higher energies ~ 160 eV at 1.17 ps [see Fig. 7(b)]. There is a significant number of low-energy electrons (<100 eV). However, the fraction of electrons in the spectrum with the energy higher than 10 keV drops very sharply by several orders of magnitude (the tail of distribution function). At 10.05 ps, the energy peak is shifted to ~ 540 eV and the energy range expands to >20 MeV [see Fig. 7(c)]. The portion of spectrum on the left-hand side with the lowest energy electrons ($<\sim 2$ eV) disappears.

There is a broad plateau in the energy distribution function of electrons extending up to $<\sim 100$ keV. During a short period from 20.09 to 23.09 ps, the energy peak of hot electrons moves from ~ 3.6 to ~ 80 keV [see Fig. 7(d) and (e)]. The electrons with ~ 100 eV are seen as the ones with the lowest energy. The right edge of the energy spectrum is expanded up to ~ 100 MeV. The fraction of electrons with energies from ~ 10 to ~ 100 MeV is negligible. At approximately ~ 25 ps, the energy peak arrives at ~ 400 keV and remains there during the rest of the PIC simulation. This 400-keV-peaked energy distribution function of electrons at 100.07 ps is illustrated in Fig. 7(f). The EEDF of electrons in the energy range from ~ 1 to ~ 200 MeV (tail of spectrum) decreases sharply by about ten orders of magnitude. For much higher laser intensities, the maximum attainable energy of ~ 500 MeV was recently reported in the PIC simulation of laser wakefield electron acceleration [43]. The fraction of electrons with energies higher than 50 MeV is negligibly small. Thus, the final EEDF of hot electrons is peaked at ~ 400 keV.

The energy spectra of electrons are also investigated at $x = 50 \mu\text{m}$. At this specific location, there is a transition

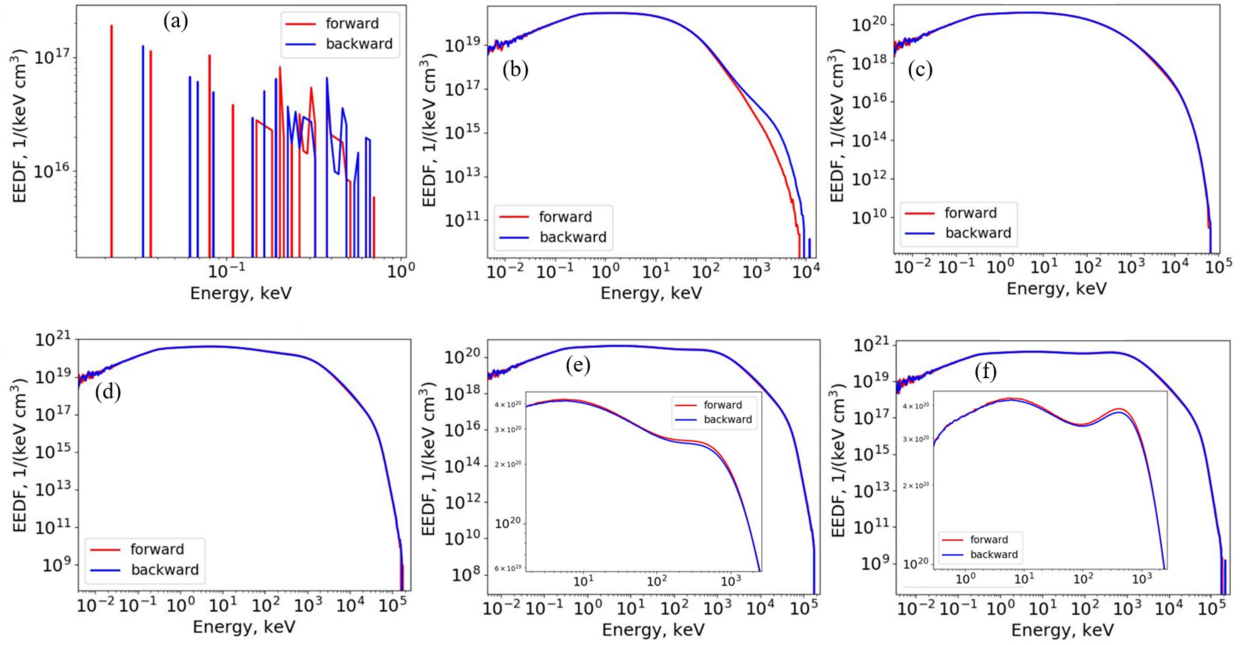


Fig. 8. Energy spectra of electrons at a screen plane placed in the center of computational domain for different times. (a) $t = 0$ ps. (b) $t = 5.09$ ps. (c) $t = 20.09$ ps. (d) $t = 50.1$ ps. (e) $t = 80.11$ ps. (f) $t = 119.9$ ps.

from the under-dense to over-dense plasma state (see Fig. 2). SMILEI has a capability to place the screen surface at any location in the computational domain and collect the energy of the macro-particles when they cross this surface. The screen plane was set perpendicular to the x -axis at point $x = 50 \mu\text{m}$ and $y = 25 \mu\text{m}$ where the electron density is equal to the critical electron density $n_{ec} \approx 9.05 \times 10^{21} \text{ cm}^{-3}$. The energy spectra of the macro-particles crossing the screen in the forward and backward x -directions at any angle were evaluated and shown in Fig. 8 for different time moments. At $t = 0$ ps, there is only a small number of thermal electrons at this location with the energies less than 1 keV which move across the screen in the forward and backward x -directions [see Fig. 8(a)]. The upper range of energy spectrum expands up to ~ 10 MeV at $t = 5.09$ ps with the energy peak located at ~ 1 keV [see Fig. 8(b)]. There is a small difference between the forward and backward tails in the energy spectrum. At $t = 20.09$ ps the peak of EEDF is shifted to ~ 5 keV and the upper range of spectrum extends to ~ 70 MeV [see Fig. 8(c)]. Both the forward and backward EEDF curves including the tails are nearly the same. At $t = 50.10$ ps, the peak near ~ 5 keV remains at its place, but the second peak near ~ 400 keV starts to develop [see Fig. 8(d)]. The 400-keV component in the electron energy spectrum becomes apparent at later times of 80.11 and 119.90 ps [see insets in Fig. 8(e) and (f)]. However, the 5-keV energy peak is still dominant. It is observed that the position of the 5-keV energy peak remains unchanged after ~ 20 ps. It should be also noticed that the distribution of electron energy is confined in the very broad 50-eV to 200-MeV range with the lower energy bound ~ 50 eV is nearly unchanged at different times. Thus, the 5- and 400-keV peaks are identified in the energy spectrum of electrons at the position of critical electron density.

IV. CONCLUSION

The 2D3V PIC simulation of Omega EP experiments is carried out using the SMILEI computer code with the goal to investigate the generation of groups of hot electrons by a laser pulse of 100-ps duration and 10^{15} W/cm^2 intensity. The population of energetic, hot electrons with the energy peak near ~ 400 keV is found to be produced due to the absorption of laser light in a plasma. The mechanisms of electron acceleration are due to the ponderomotive force and plasma instabilities. The two groups of hot electrons with the ~ 5 - and ~ 400 -keV energy peaks are locally identified in the middle of domain at a transition point from the under-dense to over-dense plasma where the electron density equals to the critical electron density. The focus of future research should be on more detailed PIC studies of physics, mechanisms, and generation processes of hot electrons and their spectra in both the under-dense and over-dense plasma.

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REPORT DOCUMENTATION PAGE

1. REPORT DATE 20230727	2. REPORT TYPE Journal Article	3. DATES COVERED	
		START DATE 20220701	END DATE 20230630
4. TITLE AND SUBTITLE			
5a. CONTRACT NUMBER HDTRA1-20-2-0001	5b. GRANT NUMBER	5c. PROGRAM ELEMENT NUMBER	
5d. PROJECT NUMBER	5e. TASK NUMBER	5f. WORK UNIT NUMBER	
6. AUTHOR(S)			
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) 1. Hopkins Extreme Materials Institute (HEMI), Johns Hopkins University, Baltimore, MD 21218, USA. 2. University of Illinois Urbana-Champaign, Champaign, IL 61820, USA 3. University of California Riverside, Riverside, CA 92521, USA. 4. University of California San Diego, La Jolla, CA 92093, USA. 5. University of Arizona, Tucson, AZ 85721, USA. 6. DEVCOM US Army Research Laboratory, FCDD-RLW-WA, Aberdeen Proving Ground, MD 21005. 7. Lawrence Livermore National Laboratory, Livermore, CA 94550, USA 8. Combustion Research and Flow Technology, Inc., Pipersville, PA 18947, USA 9. New Jersey Institute of Technology, Newark, NJ 07102, USA 10. Laboratory for Laser Energetics, University of Rochester, Rochester, NY 14623, USA. 11. Pacific Northwest National Laboratory, Richland, WA 99352, USA. 12. South Dakota Mines, Rapid City, SD 57701, USA. 13. Virginia Commonwealth University, Richmond, VA 23284, USA. 14. Washington State University, Pullman, WA 99164, USA.			8. PERFORMING ORGANIZATION REPORT NUMBER
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) DEFENSE THREAT REDUCTION AGENCY / AL-ACR 8725 JOHN J. K INGMAN ROAD FT BELVOIR VA 22060-6201		10. SPONSOR/MONITOR'S ACRONYM(S) DTRA	11. SPONSOR/MONITOR'S REPORT NUMBER(S)
12. DISTRIBUTION/AVAILABILITY STATEMENT			
13. SUPPLEMENTARY NOTES			
14. ABSTRACT			

15. SUBJECT TERMS				
16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES
a. REPORT	b. ABSTRACT	c. THIS PAGE	UU	
U	U	U		
19a. NAME OF RESPONSIBLE PERSON			19b. PHONE NUMBER (Include area code)	
Mr. Andrew Proulx, Program Manager, MSEE URA			410-516-2426	